

Quiz 8 Solution

Q1. A nearly sorted array is given to selection sort to get it fully sorted. What will be its running time?

- a) $O(n)$
- b) $O(n^2)$
- c) $O(n \log n)$
- d) $O(n^{3/2})$

Correct: b)

Q2. If a Sorted array is provided to insertion sort, how many comparisons does it make?

- a) N
- b) N^2
- c) $\log N$
- d) $N-1$

Correct: d)

Q3. Which sorting algorithm repeatedly compares adjacent elements and swaps them if they are in the wrong order?

- a) Shell Sort
- b) Insertion Sort
- c) Selection Sort
- d) Bubble Sort

Correct: d)

Q4. Which of these adjustment(s) in the original algorithm improves the Insertion Sort Algorithm?

- a) Implement Binary Search
- b) Swap elements instead of shifting
- c) Implement h-sorting
- d) All of the others

Correct: d)

Q5. Why taking powers of two in h-sorting (used in Shell Sort) typically disregarded?

- a) It increases the average-case time complexity.
- b) It may cause the algorithm to miss certain inversions, leading to poor performance.
- c) Using powers of two causes elements at even and odd indices to remain separated for too long, reducing the effectiveness of early passes.
- d) It makes the algorithm unstable for duplicate elements.

Correct: c)

Q6. Which of the following sorting algorithms has the lowest best-case time complexity?

- a) Shell Sort
- b) Insertion Sort
- c) Bubble Sort
- d) Selection Sort

Correct: b)

Q7. In which Sorting Algorithm the elements that have already been checked are in their final assigned place in the sorted array?

- a) Shell Sort
- b) Insertion Sort
- c) Bubble Sort
- d) Selection Sort

Correct: d)

Q8. Which of the following pairs of sorting algorithms are not stable by default?

- a) Bubble Sort, Insertion Sort
- b) Insertion Sort, Selection Sort
- c) Selection Sort, Shell Sort
- d) Shell Sort, Bubble Sort

Correct: c)

Q9. Which of the following best describes the relationship between Insertion Sort and Shell Sort?

- a) Insertion Sort and Shell Sort are completely unrelated algorithms.
- b) Insertion Sort is equivalent to Shell Sort, which performs only one pass with a gap of 1.
- c) Shell Sort is a repeated version of Insertion Sort, but it never reduces the gap to 1.
- d) Shell Sort never behaves like Insertion Sort, regardless of the gap size.

Correct: b)

Q10. Complete the following code for the Shell Sort Algorithm.

```
template<typename T>
void shell_sort(vector<T>& data){
    int N = data.size();

    int h = 1;
    while (h < N/3){ h = 3*h + 1; }

    while( h >= 1){
        for(int i=h; i<N; i++){
            ?
        }
        h/=3;
    }
}
```

a)

```
for(int j = i; j >= h && (data[j] < data[j-h]); j-=h){
    swap(data[j],data[j-h]);
}
```

b)

```
for(int j = i+1; j >= h && (data[j] < data[j-h]); j-=h){
    swap(data[j],data[j-h]);
}
```

c)

```
for(int j = i+1; j >= h && (data[j] > data[j-h]); j-=h){
    swap(data[j],data[j-h]);
}
```

d)

```
for(int j = i; j >= h && (data[j] > data[j-h]); j-=h){
    swap(data[j],data[j-h]);
}
```

Correct: a)

Q11. How many swaps does Selection Sort perform in the worst case on an array of size N?

- a) $N-1$
- b) $N(N-1)/2$
- c) Depends on the input
- d) $N\log N$

Correct: a)

Q12. Which of the following is true about Insertion Sort for a reverse-sorted array of size n ?

- a) It makes $n-1$ comparisons
- b) It makes $n(n-1)/2$ comparisons
- c) It makes n swaps
- d) It sorts in $O(\log n)$ time

Correct: b)

Q13. What will be the state of the array after the first full iteration ($i = 0$ pass) of Selection Sort on this array: [5, 3, 8, 1, 2]?

- a) [1, 3, 8, 5, 2]
- b) [1, 3, 5, 8, 2]
- c) [1, 3, 8, 2, 5]
- d) [1, 3, 2, 8, 5]

Correct: a)

Q14. Which of the following gap sequences is the better choice for Shell Sort?

- a) Power of 2 -1 sequence
- b) Power of 2 sequence
- c) $3x + 1$ sequence
- d) Factorial sequence

Correct: c)

Q15. Given a class Student with a public data member float gpa, which is the correct way to compare two objects s1 and s2 in a selection sort?

- a) `if(s1.gpa < s2.gpa)`
- b) `if(gpa < gpa)`
- c) `if(Student.gpa < Student.gpa)`
- d) `if(gpa1 < gpa2)`

Correct: a)

Q16. Which of the following correctly defines a generic sorting function in C++ using templates?

- a) `void sort<T>(T arr[], int n)`
- b) `template<typename T> void sort(T arr[], int n)`
- c) `sort<T>(arr, n)`
- d) `template<T> sort(arr, n)`

Correct: b)

Q17. Given the array: [8, 5, 2, 9, 1] and Shell Sort with gap sequence gap = 2, what will be the value at index 0 after the first gap-based pass?

- a) 8
- b) 2
- c) 5
- d) 1

Correct: d)

Q18. Consider the following C++ code:

```
for (int i = 1; i < n; i++) {  
    int key = arr[i];  
    int j = i - 1;  
    while (j >= 0 && arr[j] > key) {  
        arr[j + 1] = arr[j];  
        j--;  
    }  
    arr[j + 1] = key;  
}
```

What sorting algorithm is implemented here?

- a) Selection Sort
- b) Insertion Sort
- c) Bubble Sort
- d) Shell Sort

Correct: b)

Q19. Which of the following is a correct statement comparing Selection Sort and Insertion Sort?

- a) Selection Sort is adaptive; Insertion Sort is not
- b) Insertion Sort is stable; Selection Sort is not
- c) Both are unstable
- d) Insertion Sort always makes more swaps than Selection Sort

Correct: b)

Q20. What sorting algorithm does the following code implement?

```
for (int i = 0; i < n - 1; i++) {  
    for (int j = 0; j < n - i - 1; j++) {  
        if (arr[j] > arr[j + 1]) {  
            int temp = arr[j];  
            arr[j] = arr[j + 1];  
            arr[j + 1] = temp;  
        }  
    }  
}
```

- a) Selection Sort
- b) Insertion Sort
- c) Bubble Sort
- d) Shell Sort

Correct: c)